

NFL Performance Predictions

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1 Abstract

Predicting player performance in NFL games has become a critical tool for fantasy football managers, sports analysts, and bettors. Using data from nfl-data-py, this project explores the potential of a neural network regression model to forecast weekly player performance metrics such as rushing yards, passing yards, receiving yards, and touchdowns. These predictions are tailored for practical applications in fantasy football and sports betting. Using extensive play-by-play data and incorporating defensive metrics, our approach improves on traditional models, which focus primarily on offensive statistics. The results demonstrate the model’s ability to produce accurate, actionable forecasts, providing insights into its significance for real-world decision-making.

2 Introduction

Fantasy football has become a massive cultural phenomenon, generating \$30.5 billion in revenue in 2023 with projections to exceed \$114.7 billion by 2033 [2]. This growth is fueled by an ever-increasing number of participants who rely on data-driven insights to set optimal weekly lineups and gain a competitive edge. Similarly, sports betting has exploded in popularity, with millions of bettors seeking accurate predictions to evaluate player-based prop bets. The NFL, as the most popular professional sports league in the United States, has become a central focus for these industries. Thus, predicting NFL player performance has evolved into a cornerstone of modern sports analytics. Fantasy football participants and sports bettors alike rely on predictions to decide who to start in weekly lineups or determine whether players will exceed key metrics such as passing yards, touchdowns, and interceptions.

To address the demand for accurate and actionable insights, this project uses machine learning to predict quarterback (QB) performance for hypothetical future games. Using data from the open source nfl-data-py library, which offers detailed play-by-play data for every NFL game, we developed a model that forecasts key metrics such as passing yards, touchdowns, interceptions, and completion percentage. These metrics align with fantasy football scoring categories and are critical for evaluating sports betting opportunities. Our model is built on a Long-Short-Term Memory (LSTM) neural network, a deep learning architecture that excels at capturing sequential patterns and temporal dependencies. By integrating modern machine learning techniques, including embeddings, attention mechanisms, and sequence normalization, the model delivers predictions that reflect the unpredictable and dynamic nature of NFL games.

2.1 Motivation

The motivation behind this project stems from the intersection of two rapidly growing industries: fantasy football and sports betting. Accurate player performance predictions are invaluable to fantasy football participants, who can use

these insights to make start/sit decisions, navigate trades, and evaluate waiver wire pickups. For sports bettors, player-based predictions help identify valuable betting opportunities, such as yardage over/under, touchdown probabilities, or other player-specific prop bets. With more than 29 million active fantasy football participants in 2022 [3] and a booming sports betting market that saw \$16 billion bets on Super Bowl 57, the potential for data-driven decision-making is immense.

The core of this project is driven by a desire to explore the intricate relationships between game-related factors and player statistics. Quarterback performance, in particular, is influenced by a variety of contextual variables, including their recent form, the strength of opposing defenses, and situational game factors such as weather or venue. By analyzing detailed play-by-play data, this project seeks to uncover performance patterns that might not be apparent through traditional observation alone. The ultimate goal is to create a model that bridges the gap between raw NFL data and actionable insights, providing users with a competitive edge in their fantasy leagues or betting endeavors.

The effectiveness of the model will be evaluated using real-time data from the 2024 NFL season. Predictions will be compared against actual game statistics, with metrics such as Mean Squared Error (MSE) used to measure accuracy. By incorporating weekly feedback, the model will adapt to evolving conditions throughout the season, ensuring that its predictions remain relevant and reliable. This iterative evaluation process not only highlights the robustness of the model but also demonstrates its practical utility in real world scenarios.

2.2 Results

The model predicts five key quarterback performance metrics: passing yards, touchdowns, interceptions, completion percentage, and the number of times the quarterback gets sacked. These predictions provide actionable insights for a range of users, including fantasy football participants and sports bettors. Fantasy football players can leverage the model to identify quarterbacks poised to outperform or underperform expectations, optimizing weekly lineup decisions. For sports bettors, the model offers data-driven analysis for player prop bets, providing insights into yardage over/unders, touchdown probabilities, and interception risks.

The model’s effectiveness will be assessed using real-time data from the 2024 NFL season. Predictions will be compared against actual game outcomes, with accuracy evaluated using metrics such as Mean Squared Error (MSE).

3 Background Information

The NFL remains the most popular professional sports league in the United States, commanding immense fan engagement and driving significant economic impact. In 2022, over 29 million people participated in fantasy football, contributing to an \$11 billion industry [3]. By 2023, this figure had skyrocketed to

over \$30 billion [2], underscoring the immense engagement fantasy football generates for the NFL. Simultaneously, sports betting has experienced exponential growth, fueled by state-by-state legalization. For instance, betting on the Super Bowl jumped from \$7.6 billion in 2022 to \$16 billion in 2023 [1]. These trends highlight the intertwined impacts of fantasy football and sports betting on NFL engagement.

The rise of machine learning in sports analytics has further transformed how games are analyzed and decisions are made. NFL broadcasts now regularly feature machine learning models estimating probabilities for pivotal moments, such as fourth-down conversions, offering fans a data-driven lens on strategy. Our model applies similar techniques to individual player performance, helping users decide whether a player should be started in a fantasy lineup or if their projected statistics align with favorable betting odds.

The intersection of player evaluation and sports gaming has been explored in other contexts, such as Madden NFL video game ratings. However, static ratings, like those used in Madden, fail to account for real-time adaptability and nuanced changes throughout the season. This project aims to overcome these limitations by leveraging dynamic, context-sensitive machine learning tools that reflect the complexities of NFL gameplay.

The competitive landscape of fantasy football and sports betting tools has spurred significant innovation. Apps like WalterPicks compare their projections against established platforms, such as ESPN, to provide users with data-driven advice on start/sit decisions, trades, and waiver pickups. This landscape underscores the need for continual improvement in prediction accuracy and reliability. By combining extensive NFL knowledge with machine learning, our project strives to provide actionable insights that stand out in the evolving world of sports analytics.

4 Machine Learning Background

The development of our quarterback performance prediction system began with a foundational Long Short-Term Memory (LSTM) neural network architecture. This initial model was designed to process sequential game-level data for both quarterbacks and opposing defenses. Separate LSTM layers were employed to analyze quarterback and defensive histories, incorporating up to 32 games of quarterback statistics and 16 games of defensive statistics. The sequences included limited features, such as passing yards, touchdowns, air yards, sacks, and yards after the catch. Static embeddings were used to numerically represent quarterback and team identities, and the outputs of the LSTM layers were concatenated with these embeddings. The combined feature vector was passed through a series of fully connected layers to produce predictions for five core game-level metrics: passing yards, touchdowns, interceptions, completion percentage, and yards per attempt.

While the initial model provided a functional baseline for analyzing quarterback and defensive trends, it exhibited several key shortcomings. First, it

treated all games within a sequence equally, failing to emphasize the higher predictive value of recent performances. For instance, a quarterback’s performance in the last few games typically provides stronger signals for future outcomes than games earlier in the season. Without temporal weighting, the model struggled to account for evolving trends, such as improvements in a quarterback’s decision-making or adjustments in a defense’s strategy. Second, the feature set was narrow, excluding critical contextual variables such as weather conditions, home/away status, and detailed passing breakdowns by direction or depth. These omissions limited the model’s ability to account for situational nuances, such as a defense’s performance in extreme weather or against mobile quarterbacks.

Additionally, the initial model lacked an attention mechanism, which restricted its ability to identify and prioritize the most relevant portions of historical sequences. This limitation meant that all games contributed equally to predictions, even when some were less relevant due to outlier performances or changing game contexts. The architecture itself was relatively shallow, comprising only a few LSTM layers and simple fully connected layers. This restricted the model’s capacity to learn complex interactions between features. Training techniques were also basic, relying solely on the Adam optimizer with minimal regularization. Consequently, the model often overfit the training data and failed to generalize effectively to unseen scenarios.

Recognizing these limitations, we transitioned to a more advanced architecture that incorporates state-of-the-art techniques in sequence modeling. The enhanced model replaces the LSTM-based approach with a transformer-based architecture, which excels at capturing long-range dependencies in sequential data. Transformer layers utilize multi-head self-attention mechanisms to dynamically identify and prioritize the most informative parts of a sequence. For example, the model can focus on a quarterback’s recent success against strong pass rushes or a defense’s struggles with deep passing plays. By dynamically weighting sequence elements, the transformer architecture ensures that predictions are guided by the most relevant historical trends.

Another key improvement in the enhanced model is its ability to incorporate temporal weighting directly into the sequence processing. Historical data is weighted exponentially, giving greater importance to recent games while preserving the context provided by earlier performances. This approach reflects the intuition that a quarterback’s or defense’s recent form is often a stronger predictor of future outcomes than older data.

The enhanced model also benefits from a significantly expanded feature set, including advanced metrics like sack rates, deep pass rates, passing direction breakdowns, and yards per attempt. Additional contextual features, such as weather conditions, home/away status, and the quality of the receiving corps, are included to provide a more comprehensive view of the factors influencing quarterback performance. These features allow the model to account for situational variations that were previously overlooked, such as the impact of weather on deep passing accuracy or the role of a strong receiving corps in a quarterback’s efficiency.

In addition to the architectural advancements, the enhanced model incorporates more sophisticated training and regularization techniques. Batch normalization and layer normalization are applied to stabilize learning by ensuring consistent input scales across layers. Dropout regularization and gradient clipping are used to prevent overfitting and mitigate exploding gradients, enhancing the model’s robustness and ability to generalize. Identity embeddings for quarterbacks and teams have also been improved to capture latent attributes, such as a quarterback’s mobility, preferred passing depth, or a defense’s blitz frequency. These embeddings provide the model with a richer understanding of individual entities and their unique characteristics.

4.1 Model Architecture

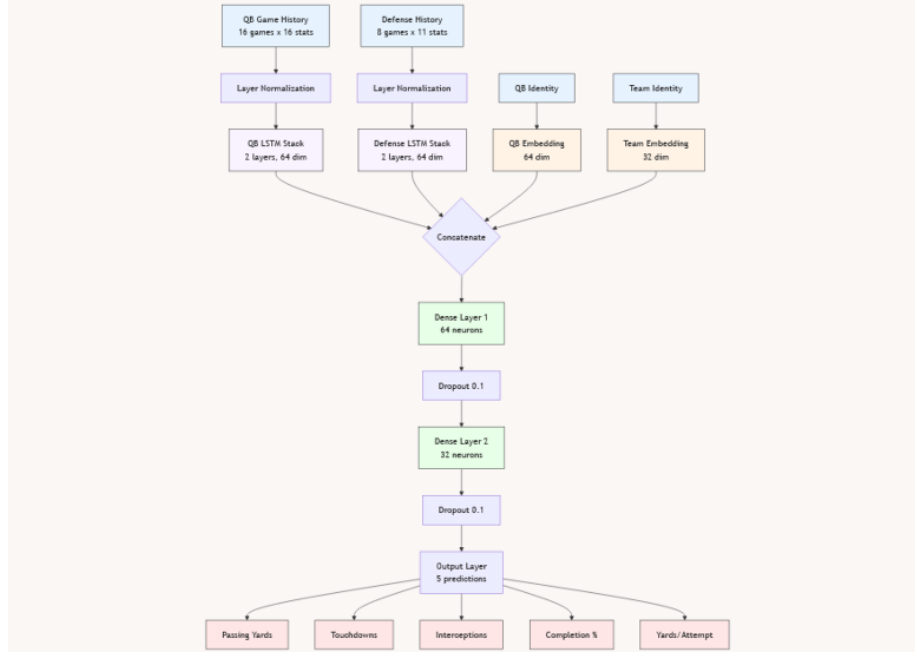


Figure 1: Model Architecture for QB Core Stats Prediction. The architecture integrates QB game history, defensive history, and embeddings for QB and team identities. Sequential data is processed through LSTM layers, while embeddings are used for static representations. The concatenated features are refined through dense layers with dropout regularization, producing predictions for passing yards, touchdowns, interceptions, completion percentage, and yards per attempt.

The initial model architecture, shown in Figure 1, was built on a foundation of Long Short-Term Memory (LSTM) networks to capture sequential trends

in quarterback (QB) and defensive history. The architecture processed up to 16 games of QB statistics and 8 games of defensive statistics, normalizing and passing them through two-layer LSTM stacks for temporal feature extraction. These sequences were complemented by static embeddings for QB and team identities, which provided numerical representations of these categorical inputs. After concatenating the LSTM outputs and embeddings, the model utilized a series of dense layers with dropout regularization to make predictions for five core quarterback performance metrics: passing yards, touchdowns, interceptions, completion percentage, and yards per attempt.

While functional, this architecture had several limitations. The use of LSTMs, although effective for sequence modeling, restricted the ability to emphasize the most relevant parts of the sequence. All historical games were treated equally, missing the higher predictive importance of recent performances. Additionally, the reliance on shallow fully connected layers and the absence of an attention mechanism limited the model’s ability to capture complex interactions between features, especially in dynamic game environments. Despite its ability to generalize to some extent, the architecture struggled with overfitting and failed to fully utilize the rich context available in the data.

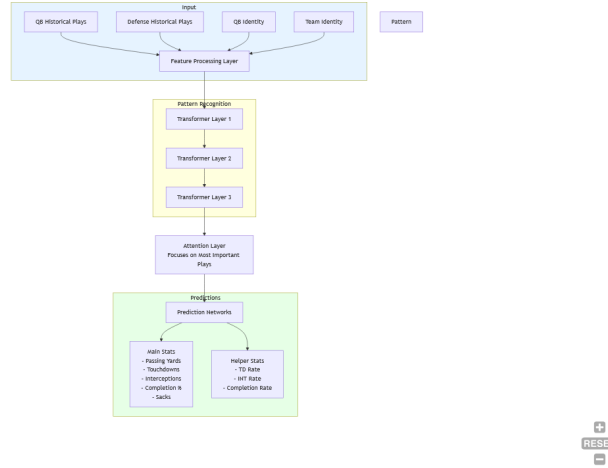


Figure 2: Enhanced Model Architecture: Transformer-Based Design for QB Core Stats Prediction. The model integrates QB and defensive play histories, embeddings for QB and team identities, and attention mechanisms. Transformer layers extract rich patterns, with outputs refined into predictions for key performance metrics.

In contrast, the enhanced model, shown in Figure 2, represents a significant evolution in the architecture by replacing LSTMs with transformer layers for sequential data processing. Transformer layers, renowned for their ability to handle long-range dependencies, are better suited for capturing nuanced rela-

tionships in sequential data. By introducing multi-head attention mechanisms, the model dynamically identifies and emphasizes the most relevant portions of the QB and defensive play sequences. This enables it to focus on critical plays or trends, such as a QB's recent performance under pressure or a defense's struggles against deep passes.

The enhanced model retains embeddings for QB and team identities but integrates them more effectively with the transformer outputs. These embeddings now interact with richer temporal features extracted by multiple transformer layers. The model's prediction network is expanded to include both the main performance metrics (e.g., passing yards, touchdowns, and interceptions) and auxiliary metrics like touchdown rate and interception rate. This additional granularity improves the interpretability of the predictions.

Notably, the enhanced architecture incorporates advanced data preparation techniques and sequence weighting. Recent games are given greater importance through exponential weighting, ensuring that predictions reflect current form while maintaining the context of longer-term trends. Additionally, the prediction network benefits from deeper fully connected layers, improved dropout regularization, and advanced normalization techniques, which enhance its robustness and generalization capabilities.

By transitioning from LSTM-based temporal modeling to a transformer-based architecture, the enhanced model demonstrates significant improvements in both predictive accuracy and flexibility. It captures richer patterns in historical data, dynamically adapts to contextual variations, and produces more precise predictions across diverse game scenarios. This evolution underscores the effectiveness of transformer architectures in tackling complex sequence prediction tasks in sports analytics.

5 Data and Analysis

The performance of the original LSTM-based model during training is illustrated in the chart titled QB Core Stats Prediction - Training History. This chart tracks both training loss (blue line) and validation loss (red line) over 40 epochs, using Mean Squared Error (MSE) as the performance metric. The training loss begins at a high value of approximately 1.3 MSE but steadily decreases as the model learns from the data, eventually reaching a minimum of 0.8098 by the 40th epoch. Similarly, the validation loss stabilizes around 1.0383, indicating effective generalization to unseen data without significant overfitting. These results highlight the effectiveness of the LSTM-based architecture in processing sequential data and balancing the model's learning from historical quarterback and defensive statistics.

Despite the solid performance of the LSTM-based model, certain trends in the training history indicate areas for improvement. For instance, while the training loss steadily decreases, the validation loss plateaus relatively early, suggesting that the model has limited capacity to further reduce error on unseen data. This limitation is likely a result of the fixed sequence length and the

inability of LSTM layers to dynamically prioritize the most relevant portions of historical data.

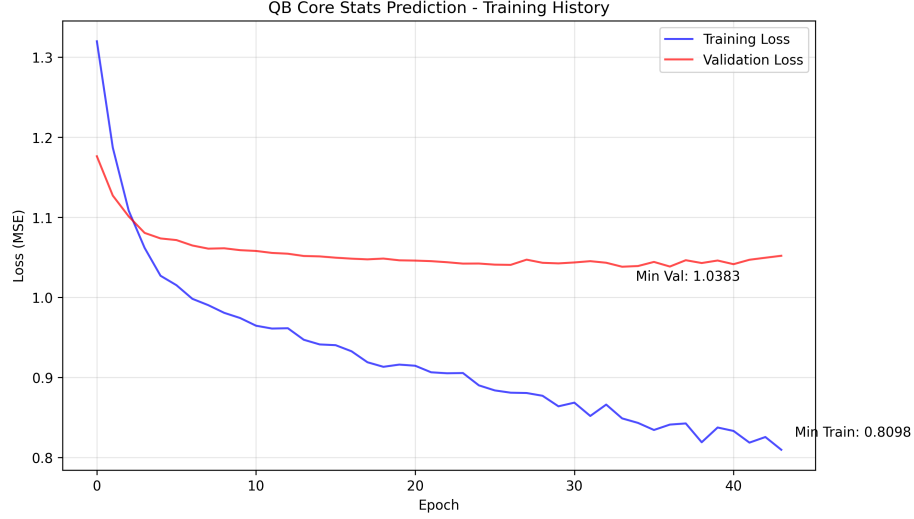


Figure 3: Training and Validation Loss Over 40 Epochs for the QB Core Stats Prediction Model. The training loss (blue) steadily decreases, reaching a minimum of 0.8098, while the validation loss (red) stabilizes at 1.0383, indicating effective learning and minimal overfitting.

In contrast, the transformer-based model demonstrates notable advancements in both training dynamics and overall performance, as illustrated in the new training history chart. The transformer model introduces a more complex architecture, incorporating multi-head attention mechanisms and the ability to process up to 2,000 plays per quarterback or defensive sequence. The training loss (solid blue line) and validation loss (solid red line) show clear improvement compared to the earlier LSTM model, with the validation loss reaching a lower minimum value of 0.8814.

Additionally, the chart tracks auxiliary losses for secondary metrics (e.g., touchdown rate and interception rate) to ensure that the model learns relationships across multiple performance dimensions. The auxiliary training loss (dotted blue line) and auxiliary validation loss (dotted red line) demonstrate consistent reductions, highlighting the transformer model's ability to process richer datasets and predict a wider range of metrics effectively.

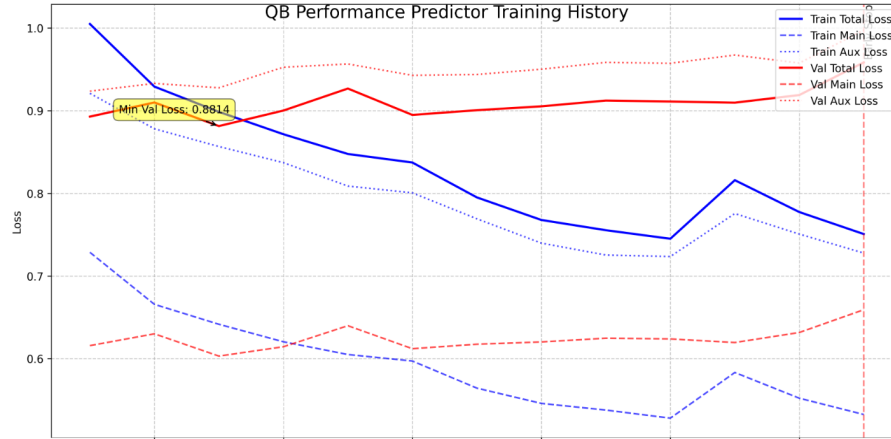


Figure 4: Training and Validation Loss for the Transformer-Based Model. The inclusion of auxiliary losses for secondary metrics demonstrates the model’s capacity to learn multi-dimensional performance relationships. Validation loss reaches a minimum of 0.8814, highlighting improved generalization.

The enhanced training dynamics of the transformer-based model can be attributed to several key improvements. By processing a larger play history of up to 2,000 plays and applying exponential weighting, the model effectively prioritizes recent data without discarding the context provided by earlier performances. Multi-head attention mechanisms allow the transformer to dynamically focus on the most relevant parts of the sequence, enabling it to uncover nuanced relationships between quarterback tendencies, defensive schemes, and game contexts.

5.1 Example Walk Through of Both Models

To illustrate how the model processes data and generates predictions, we present an example forecasting Patrick Mahomes’ performance in a hypothetical matchup against the Buffalo Bills. This walkthrough demonstrates how raw data, such as quarterback and defensive statistics, flows through the model to produce actionable predictions.

The process begins by retrieving input data for the quarterback and the opposing defense. For Mahomes, the model processes statistics from his last 16 games, such as 331 passing yards and 3 touchdowns in the most recent game. For the Bills, the model processes defensive statistics from their last 8 games, such as 245 passing yards allowed and 1 interception in their most recent game. This input data is visualized in Figures 5 and 6.

QB HISTORY (Last 16 Games)	
Game 1 (Most Recent):	331 yards, 3 TD, ...
Game 2:	298 yards, 2 TD, ...
Game 3:	285 yards, 2 TD, ...
... (13 more games)	

Figure 5: Quarterback history showing Patrick Mahomes' performance in his last 16 games.

DEFENSE HISTORY (Bills, Last 8 Games)	
Game 1:	245 pass yards allowed, 1 INT, ...
Game 2:	289 pass yards allowed, 2 INT, ...
... (6 more games)	

Figure 6: Defensive history showing the Buffalo Bills' defensive performance in their last 8 games.

Next, the raw data is normalized to ensure comparability across players and teams. For instance, Mahomes' passing yards are scaled using the formula $(331 - 280)/75 = 0.68$, where 280 is the league average and 75 is the standard deviation. This normalization process is shown in Figure 7.

2 NORMALIZATION LAYER	
QB Stats Normalized:	
Yards:	$(331 - 280) / 75 = 0.68$
TDs:	$(3 - 2.1) / 1.2 = 0.75$
... (other stats normalized)	

Figure 7: Normalization of quarterback statistics to ensure consistency across the dataset.

The normalized data is then passed through LSTM layers, which extract temporal patterns, such as Mahomes' improvement against blitz-heavy defenses

or the Bills' vulnerabilities against elite quarterbacks. The LSTM outputs are shown in Figure 8.

```
3 LSTM PROCESSING (64 dimensions)

| QB LSTM Output (example values): |
| [0.82, -0.14, 0.65, ..., 0.23] |
| |
| Defense LSTM Output: |
| [-0.31, 0.42, -0.15, ..., 0.11] |
```

Figure 8: LSTM outputs for both quarterback and defensive histories, capturing temporal patterns.

Static embeddings for Mahomes and the Bills are added next, representing unique characteristics like throwing accuracy or defensive tendencies. These embeddings are shown in Figure 9.

```
4 EMBEDDINGS

| QB Embedding (Mahomes, 64-dim): |
| [0.75, 0.82, -0.23, ..., 0.45] |
| → Represents "Mahomes-ness" |
| |
| Team Embedding (Bills, 32-dim): |
| [0.31, -0.42, 0.55, ..., -0.12] |
| → Represents "Bills Defense Identity" |
```

Figure 9: Embeddings for Patrick Mahomes and the Buffalo Bills, capturing static player and team traits.

The LSTM outputs and embeddings are concatenated into a unified feature vector, integrating temporal trends and static characteristics. This concatenated feature vector is visualized in Figure 10.

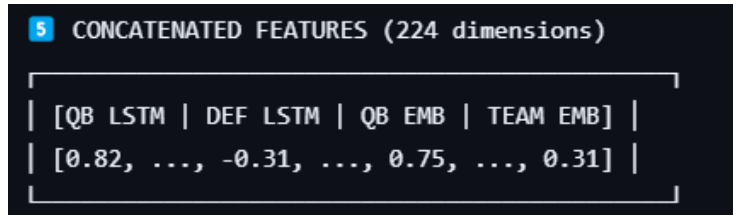


Figure 10: Concatenated feature vector combining LSTM outputs and embeddings for further processing.

The combined features are refined through dense layers to model complex interactions, such as the relationship between Mahomes' offensive tendencies and the Bills' defensive strengths. The outputs from the dense layers are shown in Figure 11.

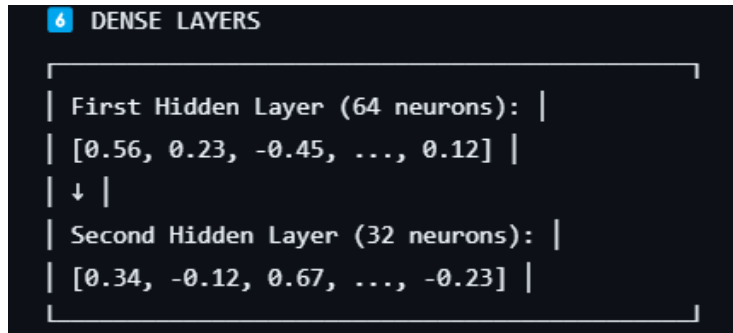


Figure 11: Dense layer outputs, refining concatenated features for final predictions.

Finally, the output layer produces predictions for key metrics, including passing yards, touchdowns, interceptions, completion percentage, and yards per attempt. These predictions are visualized in Figure 12.

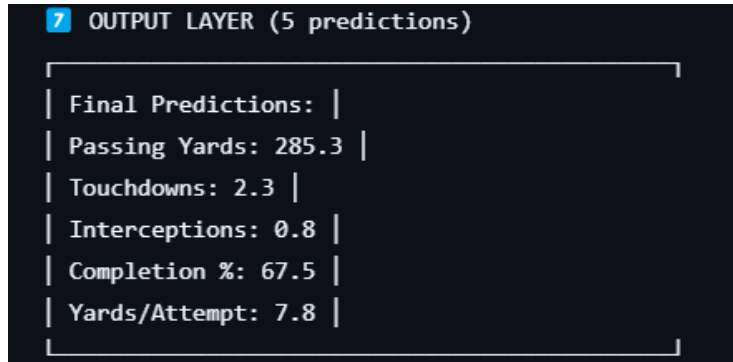


Figure 12: Output layer showing predictions for Patrick Mahomes against the Buffalo Bills.

This walkthrough highlights how the LSTM-based model integrates sequential and static data to generate actionable predictions. However, while effective, this architecture has limitations in its ability to dynamically prioritize the most relevant parts of historical sequences or fully capture complex, long-range interactions between features. To address these challenges, we developed a transformer-based model, which builds on the foundation of this architecture while introducing advanced techniques for temporal modeling and attention.

The new model processes quarterback and defensive play data in a fundamentally different and more comprehensive manner compared to the LSTM-based architecture. Instead of limiting the analysis to a fixed number of games, the transformer-based model aggregates data from up to the last 2,000 plays, capturing a larger and more detailed dataset that provides a richer context for predictions. These plays are grouped into game or week-level segments, and the model applies a dynamic weighting mechanism to prioritize the most recent performances.

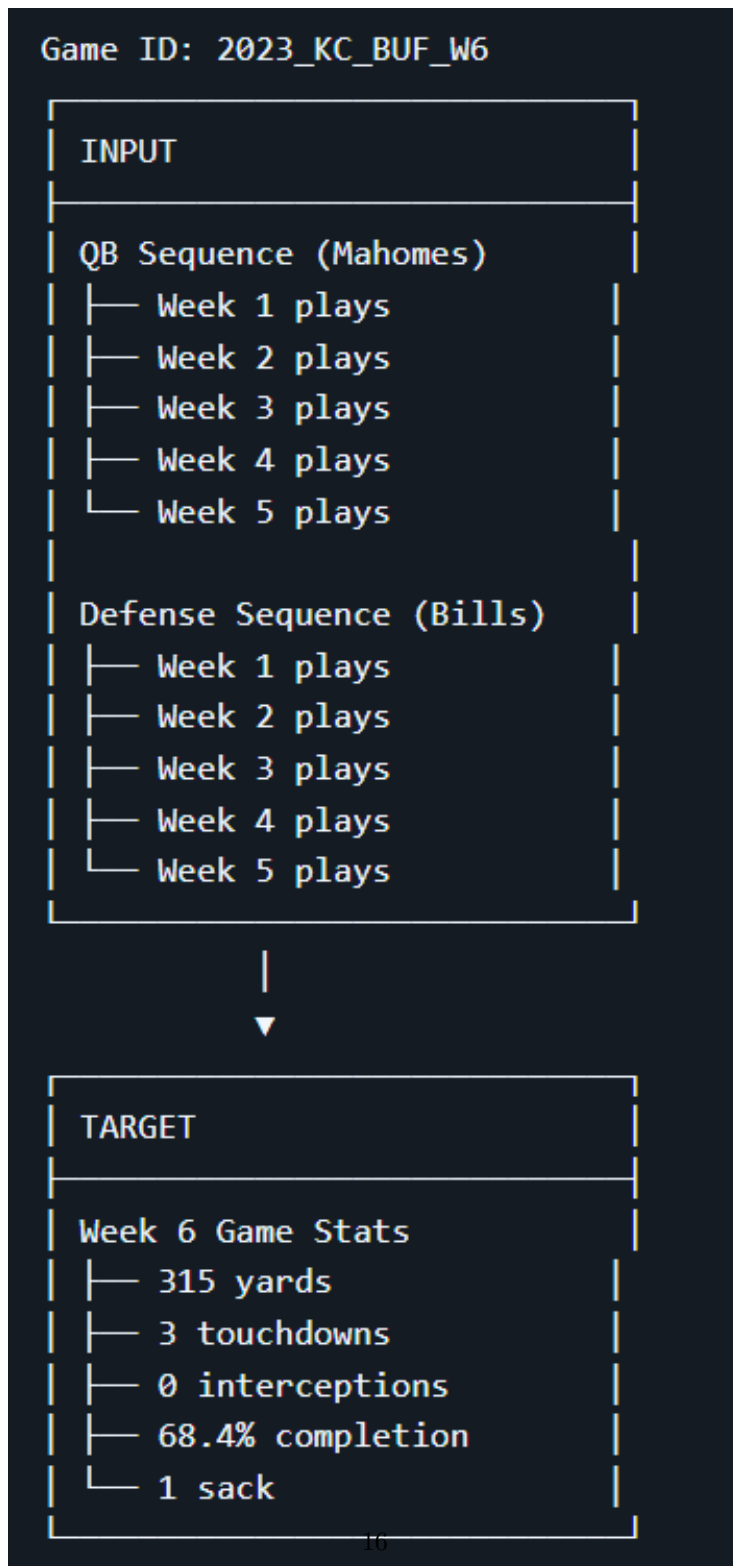


Figure 13: Transformer Model Example: Predicted Week 6 performance for Patrick Mahomes against the Buffalo Bills.

Specifically, the most recent plays are given linear weights ranging from 1.5 (most recent plays) to 1.0 (older plays), ensuring that recent trends—such as a quarterback’s improvement in form or a defense’s adaptation to new strategies—have the greatest influence on predictions. For cases where fewer than 2,000 plays are available, the remaining slots are filled with zero-weighted values, effectively neutralizing their impact on the model. This approach balances the importance of historical context with the predictive value of recent trends, while avoiding bias from incomplete datasets.

By incorporating up to 2,000 plays and dynamically weighting them, the transformer-based model offers a significant advantage in its ability to evaluate both short-term and long-term patterns. Combined with its self-attention mechanisms, this design enables the model to focus on the most relevant segments of data, ensuring that predictions are accurate, contextually aware, and robust across diverse scenarios. This enhancement addresses one of the primary limitations of the LSTM-based architecture, which treated all historical data equally and was constrained by fixed-length sequences.

5.2 Improvements

To enhance the model’s capabilities, several promising improvements are under consideration. A key direction involves adjusting the neural network to predict performance metrics for other offensive positions, including wide receivers, running backs, and tight ends. Expanding the model to these positions would enable predictions for total yards, receptions, rushing attempts, and touchdowns, offering a more comprehensive view of offensive performance and its contributing factors.

Another important enhancement is the expansion of metrics incorporated into the model. While the current version provides robust predictions for quarterback performance, integrating advanced statistics such as Expected Points Added (EPA), Win Probability Added (WPA), and passer rating would provide richer contextual insights. These metrics offer a more nuanced understanding of game dynamics, particularly in high-pressure or pivotal moments.

Additionally, there is an opportunity to include more detailed context for opposing defenses. While the model currently considers basic defensive statistics, incorporating deeper insights—such as defensive schemes, individual player performance, and coverage tendencies—could significantly improve the accuracy of predictions, particularly against complex defensive units. This would allow the model to better assess how a quarterback might perform against specific defensive alignments or strategies.

One major factor currently missing is offensive line information. Offensive line performance has a direct impact on quarterback play, influencing time in the pocket, sack rates, and overall offensive efficiency. Incorporating metrics such as pressure rates, sacks allowed, and offensive line injuries would improve the model’s ability to predict games where protection (or lack thereof) plays a critical role. This data, combined with defensive context, would enable the model to better handle scenarios with extreme outcomes, such as games with

high sack totals or exceptional offensive line performance.

Finally, these improvements collectively aim to address a limitation in the model’s current design: its difficulty in predicting “outlier games”, where specific matchups have an outsized impact on player performance. By integrating advanced metrics, deeper defensive context, and offensive line data, the model could better identify and account for games where matchups, injuries, or extreme circumstances lead to unexpected performances, whether positive or negative. These refinements would make the model not only more versatile but also more accurate in capturing the dynamic nature of football gameplay.

5.3 Limitations

One major limitation is the absence of weather conditions, such as rain and snow which are known to significantly impact gameplay. For instance, precipitation or snow fall can reduce passing efficiency, while colder temperatures often affect player stamina and ball handling. Incorporating these factors could make predictions more realistic, especially for games played in cities prone to adverse weather conditions during the later part of the season.

Another factor currently missing is the inclusion of home/away dynamics. Home-field advantage plays a substantial role in determining player and team performance, influencing crowd noise, travel fatigue, and comfort levels. Accounting for whether a quarterback is playing at home or on the road could provide critical context for predictions.

Additionally, the model does not yet integrate metrics related to formations and receiver quality, which are pivotal to understanding quarterback performance. Offensive formations, such as shotgun or play-action setups, significantly influence passing success, while receiver quality (e.g., route-running ability, separation, and yards after catch) directly impacts a quarterback’s efficiency. Including these metrics would allow the model to better contextualize quarterback performance based on the offensive system and surrounding talent.

The model also does not currently extend to different position groups, limiting its scope to quarterback performance. This narrow focus excludes insights into how running backs, wide receivers, and tight ends contribute to overall offensive success. By accounting for other positions, the model could provide a more holistic view of offensive dynamics and their effect on quarterbacks.

Furthermore, the model struggles to predict “big games” or outlier performances by quarterbacks, where they greatly exceed expectations, often due to extraordinary matchups or circumstances. These rare but impactful games are difficult to capture without deeper contextual data. Similarly, the model does not account for blowout potential, where lopsided games may lead to changes in playcalling, such as resting starters or shifting to a run-heavy strategy, which could skew predictions for key metrics like passing yards or touchdowns.

Addressing these limitations in future iterations would significantly enhance the model’s predictive capabilities, providing more contextually aware and reliable insights for both expected and unexpected scenarios.

5.4 What We Learned

Through this project, we developed a deeper understanding of LSTM architectures and RNNs while enhancing our skills in Python libraries such as Matplotlib, PyTorch, and Scikit-learn. The importance of data pre-processing and managing large datasets became especially evident during this work, as we dealt with one of the most comprehensive datasets we have ever worked with. Exploratory Data Analysis (EDA) was a particularly time-intensive process, even more so than in previous projects, due to the extensive data available through nflfastpy. The dataset spanned multiple tables, each containing dozens of columns with various types of play, game, and player data, which posed unique challenges in understanding and structuring the data.

To tackle this complexity, we created a data dictionary to document the metadata for each table and column. This helped us understand the purpose of each feature and allowed us to make informed decisions about which data to include in our model. Several factors guided our feature selection process. First, we prioritized relevance to ensure the data aligned with our project’s objectives. Second, we focused on completeness, favoring columns with minimal missing data since quarterback statistics are inherently limited by the number of games played. Finally, we identified categorical attributes and carefully decided on encoding strategies to make these features usable in our model. By organizing this metadata in the data dictionary, we streamlined the feature selection process and established a solid foundation for building the model.

This project highlighted the challenges and importance of working with large, complex datasets. Parsing through tables with dozens of attributes required a structured approach to ensure meaningful insights could be extracted. The experience emphasized the value of EDA as a critical step in machine learning workflows, as well as the need for robust pre-processing techniques to effectively manage data at scale. Overall, this process provided invaluable learning opportunities and further strengthened our skills in data analysis, pre-processing, and machine learning implementation.

6 Conclusion

In this project, we developed a machine learning model to predict NFL quarterback performance, designed to deliver actionable insights for both fantasy football participants and sports bettors. Initially built on a Long Short-Term Memory (LSTM) architecture, the model evolved into a transformer-based design that significantly enhances its ability to capture complex temporal relationships and contextual dynamics. This transformation highlights the importance of adapting cutting-edge techniques to tackle the intricacies of player performance, enabling the model to provide more precise and robust predictions.

The model integrates play-level data, embeddings, and advanced weighting mechanisms to account for recent trends and historical context. By prioritizing recent plays with exponential weighting and leveraging multi-head attention

mechanisms, the transformer architecture excels at identifying the most relevant patterns in quarterback and defensive performance. This capability ensures the model is well-suited for real-world applications, such as optimizing fantasy football lineups and evaluating player-based prop bets. The growing popularity of these industries underscores the demand for accurate player performance forecasts, and this project demonstrates how machine learning can address that demand.

Despite its strong performance, the model has limitations that highlight opportunities for further improvement. Currently, the absence of weather conditions, home/away factors, and detailed receiver metrics constrains the model’s ability to account for all key variables influencing player performance. Additionally, the model struggles to predict extreme outlier games or scenarios involving blowouts, where unique matchups or in-game dynamics lead to unexpected results. Expanding the model to include advanced metrics like Expected Points Added (EPA), Win Probability Added (WPA), and offensive line performance would enhance its predictive power and contextual awareness. Furthermore, adapting the architecture to predict performance for other offensive positions—such as wide receivers, running backs, and tight ends—could broaden its scope and utility, making it a comprehensive tool for analyzing team dynamics.

The transition from the LSTM-based model to the transformer architecture underscores the importance of innovation in tackling complex problems in sports analytics. The transformer model’s ability to dynamically prioritize relevant data and incorporate richer feature sets represents a significant leap forward in predictive accuracy and flexibility. This evolution demonstrates how machine learning can bridge the gap between raw NFL data and actionable insights, empowering users to make informed decisions in two rapidly growing industries.

Looking ahead, we aim to refine the model further by addressing its current limitations and incorporating additional features that capture the nuanced factors influencing NFL games. These enhancements will ensure the model remains relevant and reliable in dynamic scenarios, such as late-season games, playoffs, and extreme weather conditions. By continuing to evolve the architecture and expand its scope, this project demonstrates the potential of machine learning to revolutionize sports analytics, empowering fantasy football enthusiasts and sports bettors with cutting-edge, data-driven tools.

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